L2b: d_reg trajectory of the constrained prime-field Semaev system

(incomplete) **Lead.** L2b of `research/notes/ecdlp-general/RESEARCH_PRIME_FIELD_LEADS_20261031.md`; criteria frozen in `PREREG.md` before the ladder (reopens: real-arm ratio of the median maximal step degree < 1.5 while the control ratio ≥ 2; null_confirmed: the two ratios agree within 25 %). **Status.** incomplete: the frozen ladder (`run_ladder.sh`: k = 10, 12, 14, 16 with 2 curves × 4 targets × 2 arms, k = 18 and 20 reduced) produced one cell before the host stopped the workflow. **Decision.** none; no trajectory ratio can be formed from one cell.

This report was written by the synthesis step (2026-11-01) from `results/cells.csv`, `results/instances.jsonl`, `results/probe_cells.csv`, `results/probe_instances.jsonl`, `PREREG.md`, `run_cell.sh`, `run_ladder.sh` and the generator `gen/src/main.rs`. No number is new, and nothing was rerun.

1. Instrument

Generator `gen` (Rust) writes, for (p, curve seed, target seed, arm), the system $S_4(X_1, X_2, X_3, x(R)) = 0$ plus three membership polynomials of degree N = round(p^{1/3}) (real arm: the interval base; control arm: a random polynomial f of the same number of terms and total degree with random membership polynomials of the same degree). msolve (`-g 2 -v 2 -t 1`) under a watchdog (time cap and 16 GiB RSS cap). Metrics parsed from the F4 round table: maximal step degree, last degree, rounds, largest matrix, last matrix, total rows, basis size, and `zdeg` as a stated proxy only (not a first fall degree). Wall time is used only as the kill cap.

2. Every cell attempted

Ladder (`results/cells.csv`, seeds from the frozen ladder):

p	$\log_2 p$	N	arm	status	max deg	last deg	rounds	largest matrix	total rows
65521	16.0	40	ctrl	ok	48	47	126	14235 × 14229	314140

Probe cells (`results/probe_cells.csv`, real arm, curve and target seed 9, outside the frozen ladder's seeds; run to check the generator and caps before the ladder, cap 600 s):

p	$\log_2 p$	N	arm	status	max deg	last deg	rounds	largest matrix	total rows
4093	12.0	16	real	ok	24	23	273	2059 × 2033	39533
16381	14.0	25	real	ok	33	32	585	6093 × 6099	175479
65521	16.0	40	real	ok	48	47	1629	17516 × 17418	1444706
262139	18.0	64	real	timeout 600 s	72 (partial)	48	2119	7279 × 31831	7462060
1048573	20.0	102	real	timeout 600 s	110 (partial)	106	317	18822 × 81222	3895033

Observed, not decided: in every completed cell, real or control, the maximal step degree is N + 8 (24, 33, 48, 48), which is the membership degree plus a constant, as `PREREG.md` said to expect by

construction ("both arms contain three univariate membership polynomials of degree N, so max deg ≥ N in both arms regardless of S_4"). The two timed-out cells reached degree N + 8 as well before the cap. At p = 65521 the one control cell and the one (different-seed) real probe cell agree in maximal degree (48) and in last degree (47) and differ in rounds (126 against 1629) and total rows ($3.1 \cdot 10^5$ against $1.4 \cdot 10^6$); with one cell per arm and different seeds this is not a comparison.

Graph: `dreg_vs_log2p.svg` (maximal step degree and largest matrix width against $\log_2$ p for the six recorded cells, hollow markers for the two killed cells; drawn by `plot.py` from the CSV files, with `results/summary.csv` from `summarize.sh`).

3. Decision against the frozen rule

Neither clause can be evaluated: the trajectory ratio needs both arms at the largest p where both completed every run and at p = 1021, and the ladder has one arm at one p. Status incomplete; no decision; the lead card's question stays open with the instrument built and the criteria frozen.

4. What a rerun needs

The ladder as written (`PAR=8`, cap 1500 s per cell) on a quiet host; the k = 18 and k = 20 cells will exceed 600 s on the real arm (both probes timed out) and are the ones the reopen clause depends on. Memory cap 16 GiB was not reached in any cell.

5. Limits

- Design (the metric is pinned to N + O(1) by the membership polynomials) was acknowledged in `PREREG.md`; a flat real-arm trajectory in maximal step degree is not expected from the degree profile alone, so the matrix profile is the quantity that would carry a decision.
- No PDF renderer was available to the lead agent; `REPORT.pdf` was rendered by the synthesis step with Typst in a scratch virtualenv.

Rerun 2026-11-01: the frozen ladder, complete (appended; the sections

above are unchanged) Plan `../rerun_20261101/RERUN_PLAN.md`, report `../rerun_20261101/REPORT.md`, data `../rerun_20261101/L2b/`. msolve 0.6.5 with the exact sparse/dense kernel (`-l 1`), which reproduces the three completed probe cells above and the one ladder cell in every column. The ladder of `run_ladder.sh` ran as listed: 70 cells, 68 completed, the two p = 1048573 cells stopped at the 1500 s cap.

Largest p where both arms completed every run: 262139. Median maximal step degree 18 at p = 1021 and 72 at p = 262139 in both arms: trajectory ratio 4.00 (real) and 4.00 (control). **Decision: null_confirmed.** Every completed cell has maximal degree N + 8. Observed, not decided: the real arm does more F4 work than the control at every rung (median total rows 1.26 to 21.6 times the control's from p = 1021 to 262139).